

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 1

Adaptive Classified

Chapter 04 · Inequalities

Questions

Student book · questions with writing space

- 16 pages
- Past papers up to June 2026

Inequalities

Your classified route

6

questions

41

marks

2

families

6

variants

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CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

Inequalities

Families and variants

F01 • One-variable Inequalities

V01 Factorable Quadratic Sign Inequality

V02 Rational Inequality with Critical Points

V03 Discriminant Condition to a Parameter Range

F02 • Graphical Inequality Regions

V01 Shade a Given System of Inequalities

V02 Write a System from a Shaded Region

V03 Derive Boundary Constants, then Write the Region

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One-variable Inequalities

WHAT THIS FAMILY ASKS

Reduce an inequality to factors or a quotient, identify all critical values and use sign or graph logic to state the complete solution set.

VARIANT ROUTE

V01 Factorable Quadratic Sign Inequality

V03 Discriminant Condition to a Parameter Range

2 Complete Classified questions

10 total marks

1.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

Solve the inequality

$$4x^2 - 3x + 7 \geq 4x + 9$$

(4)

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8. The straight line l has equation $y = k(2x - 1)$, where k is a constant.

The curve C has equation $y = x^2 + 2x + 11$

Find the set of values of k for which l does not cross or touch C .

(6)

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Graphical Inequality Regions

WHAT THIS FAMILY ASKS

Translate between systems of linear/quadratic inequalities and their coordinate-plane regions, including deriving boundary constants or intersections when necessary.

VARIANT ROUTE

V01 Shade a Given System of Inequalities

V02 Write a System from a Shaded Region

V03 Derive Boundary Constants, then Write the Region

4 Complete Classified questions

31 total marks

3. (i) Find the range of values of z for which

$$\frac{4}{z} < 8$$

giving the answer in simplest form.

(2)

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(ii)

Figure 1 shows a plot of

- the line with equation $y = x + 5$
- the curve with equation $y = x(x - 3)$

On the opposite page there is a copy of Figure 1 labelled Diagram 1.

On Diagram 1 represent the following inequality graphically.

$$x + 5 \leq y \leq x(x - 3) \quad \text{for } x \geq -2$$

(3)

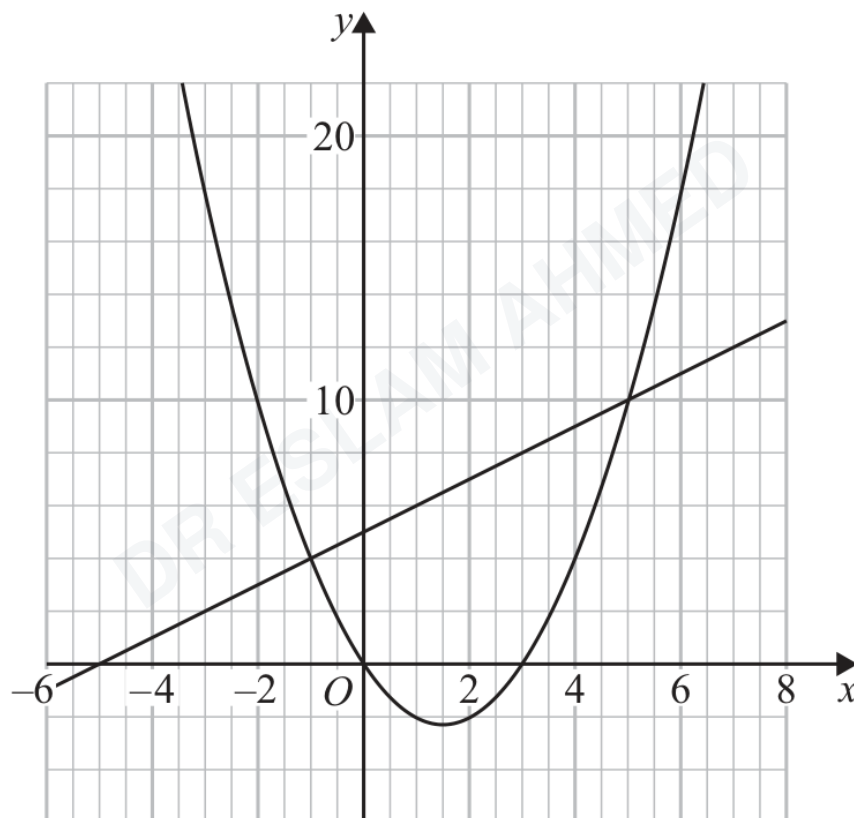
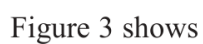


Diagram 1

In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.

- the line l with equation $y - 5x = 75$
- the curve C with equation $y = 2x^2 + x - 21$

(a) Find, using algebra, the coordinates of P and the coordinates of Q .



- The line l intersects the curve C at the points P and Q , as shown in Figure 3.

The region R , shown shaded in Figure 3, is bounded by C , l and the x -axis.

(b) Use inequalities to define the region R .


$$0 \leq y \leq 2 \qquad y \leq 10 - 2x \qquad y \leq kx$$

The line $x = a$, where a is a constant, passes through the intersection of the lines $y = 2$ and $y = kx$

Given that the area of R_1 is $\frac{27}{4}$ square units,

(a) find

(i) the value of a

(ii) the value of k

(4)

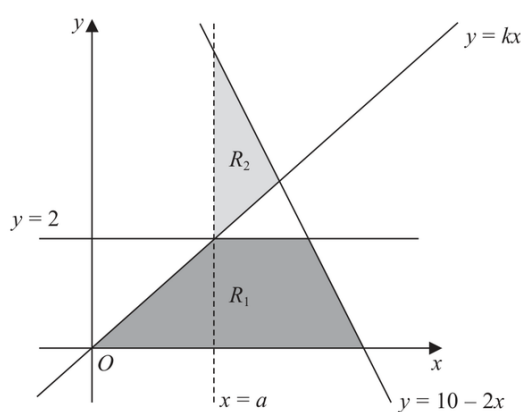


Figure 2

(b) Define the region R_2 , also shown shaded in Figure 2, using inequalities.

(2)

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9.

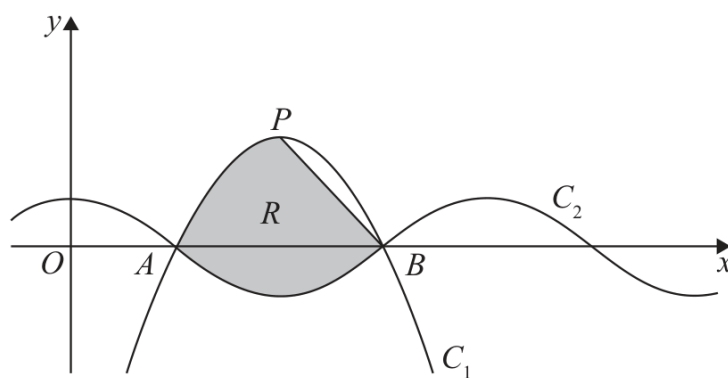


Figure 2

- (a) Express $6x - \frac{27}{4} - x^2$ in the form $a + b(x + c)^2$ where a , b and c are constants to be found.

(3)

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The figure shows a Cartesian coordinate system with a horizontal x-axis and a vertical y-axis. The origin is labeled O. Two curves, labeled C_1 and C_2 , intersect at two points on the x-axis, labeled A and B. Curve C_1 is concave down, and curve C_2 is concave up. The region between the two curves from A to B is shaded and labeled R. A point P is marked on the boundary of region R, specifically at the highest point of the upper curve C_2 within the interval [A, B].

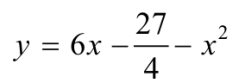
Figure 2 shows part of a sketch of curve C_1 with equation

$$y = 6x - \frac{27}{4} - x^2$$

Given that the point P is the maximum point on C_1

(b) state the coordinates of P

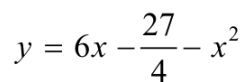
(2)



where k is a constant and x is measured in radians.

(c) (i) state the x coordinate of B

- (ii) state the value of k
- (iii) state the period of C_2



The line segment L joins P and B .

(d) Use inequalities to define R .

(5)